QoE-Aware DRL for Slice Admission Control: A Live Open RAN Testbed Validation

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***Abstract*—Building on our previous work on Deep Reinforcement Learning (DRL)-driven Slice Admission Control (SAC), which was validated entirely in simulation, this paper moves the same control problem onto a real Open RAN testbed. We deploy a Quality of Experience (QoE)-aware Deep Q-Network (DQN) admission policy on a real OpenAirInterface (OAI) gNodeB (gNB) across three network slices (eMBB, URLLC, mMTC), and compare it against a static baseline and a best-possible static-at-cap policy. After identifying and correcting a calibration issue in our QoE estimation, a fully powered live re-evaluation (46 episodes per arm) shows DQN under the QoE-aware reward sustains 100% SLA compliance throughout, while the SLA-only reward and the static-at-cap arm converge to a statistically indistinguishable rate (44 of 46 episodes fully compliant for each, Fisher exact $p = 1.0$), both modestly above the static baseline's 42 of 46: the collapse-avoidance edge our earlier, smaller-sample evaluation suggested for the SLA-only reward does not hold at this larger, well-powered scale, though the QoE-aware reward's does. We additionally live-evaluate checkpoints trained for a congested, URLLC-preservation scenario: unlike its offline counterpart, where DQN measurably trades eMBB compliance for URLLC under an imposed shared-resource-pool constraint, every arm is fully SLA-compliant live, since this rig's calibrated ceilings are too small relative to the gNB's real capacity for the same forced scarcity to arise – though the QoE-aware reward still yields a materially higher URLLC Mean Opinion Score at the same compliance level. We also report the control-loop latency and model footprint needed for practical deployment. These findings demonstrate the feasibility, and surface the honest limits, of deploying DRL-driven SAC on real Open RAN hardware.**

***Index Terms*—5G, Network Slicing, Slice Admission Control, Deep Reinforcement Learning, Quality of Experience, Open RAN**

I. Introduction and Background

Network slicing (NS) lets a single physical 5G deployment serve heterogeneous service classes under per-slice Service Level Agreements (SLAs): Ultra-Reliable Low-Latency Communication (URLLC) requires sub-millisecond latency and high reliability, Enhanced Mobile Broadband (eMBB) demands high throughput, and Massive Machine-Type Communication (mMTC) accommodates large numbers of low-power devices. Slice Admission Control (SAC) decides whether an incoming slice request can be accommodated given current resource availability. Static, threshold-based SAC cannot adapt its allocation as demand shifts, motivating context-aware, learning-based approaches.

Our prior work applied Deep Reinforcement Learning (DRL) to SAC under gNodeB (gNB) congestion, using SLA-driven reward shaping to prioritise URLLC traffic, and later extended this to a third slice (mMTC) with a load-balancing objective across multiple gNBs [1], [2]. Both studies, along with the majority of DRL-driven SAC work surveyed in our systematic review of the 2024–2026 literature [3], were validated entirely in simulation. This paper closes that gap for our own formulation: we realise the same admission-control problem on a real OAI gNB over its native E2 control interface, with the radio channel still simulated (rfsim), and centre the evaluation on Quality of Experience (QoE) rather than SLA compliance alone.

This paper makes the following contributions:

1) We deploy a DRL-based SAC policy on a real O-RAN E2 control loop and identify a structural difference from our earlier simulated work: the real E2 interface exposes no per-flow accept/reject primitive, only a per-slice resource ceiling, so admission must be realised as an adjustment of that ceiling.
2) We compare a QoE-aware reward directly against the SLA-only reward of [1], [2] on the same live hardware.
3) We add a static-at-cap arm that isolates whether the learned policy's advantage is genuine admission timing, or simply the result of using a more generous static allocation.
4) We report the engineering numbers (control-loop latency, model size) a real deployment needs, a calibration issue found and fixed in our QoE estimation, and the fully powered re-evaluation that correction enabled.
5) We evaluate a congested, multi-slice traffic scenario both offline and, at a smaller scale, live, showing where the offline scarcity mechanism does and does not carry over to real hardware.

The remainder of this paper is organised as follows: Section II reviews related work on DRL-driven SAC. Section III describes the live testbed, the problem formulation, and the reward function. Section IV presents and discusses the results.

Section V concludes the paper and outlines future work.

II. Literature Review

TODO(MANOJ): Literature review to be added here, following the style of Section II in our prior conference papers [1], [2] (narrative summary of ∼10–12 related DRL-driven SAC / real-testbed studies, each introduced with "The authors in [X] proposed..."), concluding with a short paragraph positioning this paper against that related work. Candidate sources are already screened in our systematic review [3].

III. Methodology

A. Testbed

We deploy a real OAI gNB (ORANSlice fork [4]) with a simulated radio channel (rfsim), an Open5GS core, and three `nr-uesoftmodem` user equipment (UE), one per slice. The gNB has a nominal capacity of 106 Physical Resource Blocks (PRBs, $\mu = 1$ numerology). Unlike our earlier simulated work, the real O-RAN E2 interface does not expose a binary accept/reject action; its only control primitive is a per-slice ratio ceiling on a 0–100 scale, enforced by the MAC scheduler [5]. We therefore realise the admission decision as a ceiling adjustment: an accept decision raises a slice's ceiling toward a calibrated maximum, and a reject decision lowers it toward a floor. Ceiling values were calibrated by direct measurement on the live rig rather than assumed, since the 0–100 ratio scale does not map linearly onto real PRBs at this cell's configuration. The static baseline holds its ceiling fixed for the whole episode; a second static arm, *static-at-cap*, instead pins every slice's ceiling at its calibrated maximum from the start, isolating whether a more generous fixed allocation alone can match a learned policy. Learned arms are DQN policies [6], trained offline against a closed-loop simulated traffic source, then evaluated live with frozen weights.

B. Problem Formulation

The gNB state at time t is defined as

$$s_t = \left(\frac{U_s(t)}{B},\ C_t,\ \frac{L_s(t)}{L_{\max}}\ \forall s \in \mathcal{S} \right) \tag{1}$$

where $U_s(t)$ is the resource allocation of slice $s \in \mathcal{S} = \{\text{eMBB}, \text{URLLC}, \text{mMTC}\}$, B is the total resource budget on the ceiling's normalised 0–100 scale, C_t is the current congestion level, and $L_s(t)$ is slice s's queue length, normalised by its maximum capacity $L_{\max}$. The agent's action at each step is the ceiling adjustment described above. The reward function is

$$r(s_t, a_t) = \sum_{s \in \mathcal{S}} \big(\omega_s R_s - \lambda_s\big) - \mu C_t \|a_t\|_1 + \eta\, \text{MOS}_t \tag{2}$$

where ω_s and λ_s are slice s's reward weight and SLA-violation penalty, R_s is the reward for serving slice s, and $\mu C_t \|a_t\|_1$ penalises total accepted volume by the current congestion level, matching the SLA-driven reward structure of our earlier work [1], [2]. Setting the QoE weight $\eta = 0$ recovers the SLA-only reward used in that prior work; setting $\eta = 1$ adds a Mean Opinion Score (MOS) term, estimated by a calibrated QoE-scoring model, giving the QoE-aware reward evaluated here. Both variants are compared on the same live hardware. We also report a priority-weighted compliance score, using each slice's declared importance weight rather than an unweighted average across slices of unequal importance.

TABLE I
Live SLA compliance (%), 46 episodes/arm

Arm	eMBB	URLLC	mMTC
Static baseline	91.7	91.4	91.4
Static-at-cap	97.4	95.7	95.7
DQN (SLA reward)	96.6	95.7	95.8
DQN (QoE reward)	**100.0**	**100.0**	**100.0**

C. Evaluation Setup

Each of the four arms was evaluated live across 11 random seeds, with frozen policy weights: an initial batch of 3 seeds at 2 episodes each, extended in two further rounds of 3 and 5 seeds at the full 5-episode protocol once a QoE-calibration issue discovered during this study (Section IV-B) was corrected, for 46 fully-logged episodes per arm – large enough that a Fisher exact test between two arms tied at their observed rate is a genuine null result, not merely an underpowered one. The static-at-cap arm reuses the static baseline's own control code unmodified, with its ceiling parameter changed only in configuration. Training was performed offline, over 300 episodes per seed, against a closed-loop simulated traffic source before live deployment.

IV. Results and Discussion

A. Live Testbed Results

Table I summarises SLA compliance for all four arms across 46 live episodes each, following a fully powered re-evaluation under the corrected QoE calibration described in Section IV-B. DQN under the QoE-aware reward sustains 100% SLA compliance on every slice, in every episode (46/46 episodes fully compliant). DQN under the SLA-only reward and the static-at-cap arm reach an identical, slightly lower rate (44/46 each); the static baseline trails both at 42/46. Fig. 1 illustrates the underlying mechanism: the baseline's commanded ceiling stays flat at its starting value for the whole episode, while DQN raises each slice's ceiling toward its calibrated maximum within the first few steps and holds it there.

A natural question is whether DQN's advantage simply reduces to using a more generous fixed ceiling. Our earlier, smaller-sample evaluation (15 static-at-cap and a reverified 25 DQN-SLA episodes) suggested a clear answer: static-at-cap collapsed in 27% of episodes while DQN-SLA never collapsed once (Fisher exact test, $p = 0.0149$). At this larger, fully powered sample, that edge does not hold for the SLA-only reward: DQN-SLA and static-at-cap reach an identical 44/46 fully-compliant episodes (Fisher exact $p = 1.0$) – a large enough sample that this is a genuine null result, not merely insufficient power. Both do, however, clearly separate

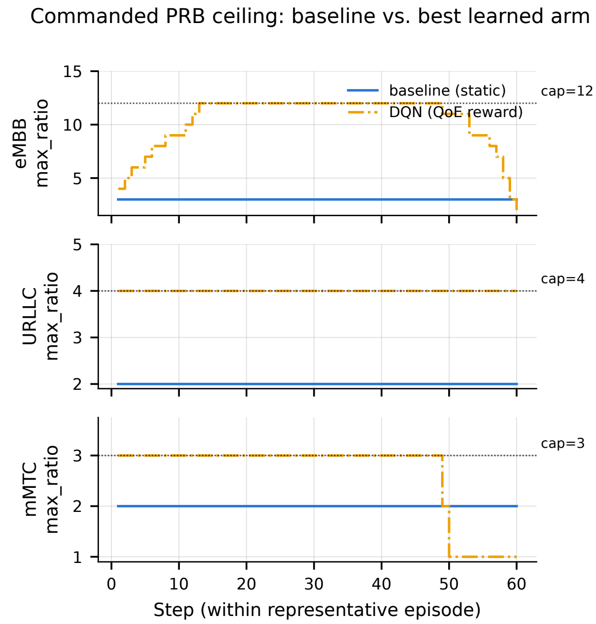

Fig. 1. Commanded ceiling per slice over one representative episode: static baseline (flat) vs. DQN (rises to its calibrated maximum).

from the static baseline's 42/46, though not to a statistically significant degree at this sample size (Fisher exact $p = 0.68$ for each vs. baseline). The QoE-aware reward is the one variant whose advantage is unambiguous, with a clean 46/46 record throughout – though even its comparison against the baseline is not itself statistically significant here (Fisher exact $p = 0.12$), since the baseline's own compliance is high enough at this scale to leave less room to separate from. We report both rounds rather than only the more favourable one: a fixed allocation's failure mode is real and shared by every non-learning arm, but the SLA-only reward's learned admission timing does not distinguishably improve on a well-chosen fixed ceiling here, contrary to what our initial, smaller sample suggested; only the QoE-aware reward's advantage is unambiguous, and even that awaits a larger sample to separate cleanly from the baseline.

Fig. 2 shows this episode-by-episode: DQN under the QoE-aware reward sits at exactly 100% in every episode, while the static baseline, the static-at-cap arm, and DQN under the SLA-only reward each show the same bimodal split between fully-compliant and collapsed episodes.

B. Deployment Feasibility

Measured directly against the real gNB, the control loop's E2 message round trip has a median latency of 0.57 ms, and the DQN model itself (under 20,000 parameters) selects an action in under 70 μs on CPU. Both are a small fraction of the 5-second control interval used in this study, indicating the learned policy adds negligible overhead to the control loop.

Comparing offline training predictions against the live results showed a good match for eMBB traffic, but not for URLLC and mMTC, which were trained under a different demand assumption than what was observed live. During

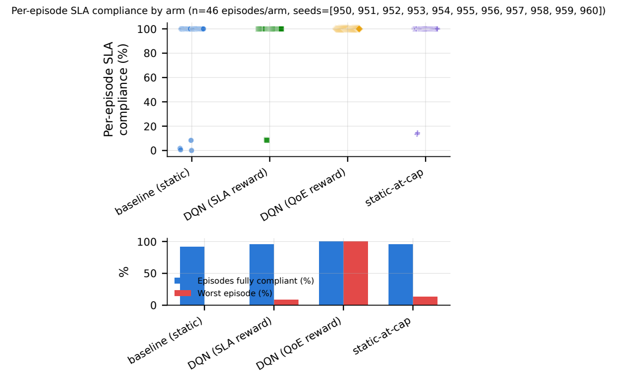

Fig. 2. Per-episode SLA compliance (%). Top: one point per episode. Bottom: fraction of episodes fully SLA-compliant and worst single-episode compliance. Baseline, static-at-cap, and DQN under the SLA-only reward all show the same bimodal split; only DQN under the QoE-aware reward stays uniformly compliant.

TABLE II
SLA COMPLIANCE (%) UNDER CONGESTED, RANDOMISED TRAFFIC (OFFLINE), 165 EPISODES/ARM

Arm	URLLC	eMBB	mMTC
Static baseline	19.7	**32.7**	**19.8**
DQN (SLA reward)	**27.7**	7.7	9.4
DQN (QoE reward)	25.0	8.2	11.6

this comparison we also identified a calibration issue in our QoE estimation: the throughput measure used to estimate MOS during training did not match the units produced by the live environment, which understated MOS for well-served traffic. We corrected this calibration and verified it against real observed operating points; Table I above reports the resulting fully powered live re-evaluation.

C. Congested Scenario (Offline)

Sections IV-A and IV-B use one constant demand level per episode. To test a harder, more realistic scenario, we evaluate the same trained policies offline under continuously varying, randomised multi-slice traffic with genuine cross-slice resource contention: when combined demand exceeds the shared PRB pool, every slice's served allocation is scaled down in proportion to its request, rather than each slice being served independently. Table II reports SLA compliance under this scenario, pooled over 11 seeds (165 held-out episodes per arm).

Under both reward variants, DQN raises URLLC compliance above the static baseline (19.7% $\rightarrow$ 27.7% for the SLA-only reward, $\rightarrow$25.0% for the QoE-aware reward) by actively trading off eMBB and mMTC compliance – the same URLLC-first behaviour our earlier simulated work targeted [1], [2], now shown under real cross-slice contention rather than each slice's demand being satisfied independently. Unlike the SLA-only reward's live collapse-avoidance edge (Section IV-A), this tradeoff replicates cleanly on an independent, 8-seed extension of this same evaluation, with only modest quantitative shifts from the initial 3-seed round. This tradeoff, however, is not a net win once each slice's declared importance is taken into account: weighted by each slice's priority, the static baseline still scores higher overall (24.9% vs. 19.1% for DQN with the SLA reward and 17.9% with the QoE reward), because eMBB's declared importance is high enough that the compliance it gives up outweighs URLLC's gain. The tradeoff is real and repeatable, but whether it or the underlying priority weights should be revisited is an open question for future work.

We additionally live-evaluated the same congested/URLLC-preservation checkpoints on the real rig, at a smaller pilot scale (one seed, 2 episodes per arm). Unlike the offline result above, every arm – including the static baseline – was fully SLA-compliant live. This is a scale mismatch, not a contradiction: the offline scenario's tradeoff is driven by an imposed shared-resource-pool budget deliberately set smaller than what the three slices' ceilings could jointly request, whereas this rig's calibrated ceilings (summing to a small fraction of the gNB's real 106-PRB capacity) never approach genuine physical contention at these traffic levels, so no forced tradeoff arises live. The QoE-aware reward nonetheless produced a materially higher URLLC Mean Opinion Score (4.83) than the SLA-only reward (2.44) or the static baseline (2.66) at the same 100% compliance – a difference in experience quality the raw compliance numbers alone do not show. A live reproduction of genuine physical multi-slice scarcity would require substantially higher per-slice ceilings and traffic than this campaign's calibration, which we leave for future work.

D. Summary of Findings

At full statistical power, the QoE-aware reward is the arm whose advantage over static control clearly holds up: a clean 46/46 fully-compliant record against 44/46 for DQN-SLA and static-at-cap and 42/46 for the static baseline. The SLA-only reward's own edge over a well-chosen fixed ceiling, which an earlier, smaller-sample round suggested was equally clear, does not replicate here: DQN-SLA and static-at-cap are statistically indistinguishable at this sample size. The offline congested scenario shows the same policies can learn to prioritise URLLC under harder, more realistic contention, consistent with our earlier simulated findings, though this is not yet an unambiguous improvement once every slice's relative importance is accounted for, and does not reproduce live at this campaign's ceiling scale. Together, these results narrow the gap between our earlier simulation-only studies and a real deployment, while surfacing an honest, more nuanced picture than a smaller sample alone would have shown.

V. Conclusion and Future Work

This paper realised the DRL-based slice admission control problem from our prior simulated work [1], [2] on a real OAI gNB over its native E2 control loop, with the radio channel still simulated. A calibration issue in our QoE estimation was identified and corrected, enabling a fully powered live re-evaluation (46 episodes/arm): DQN under the QoE-aware reward sustained 100% SLA compliance throughout, while the SLA-only reward and static-at-cap converged to an identical, statistically indistinguishable 44/46 rate, both modestly above the static baseline's 42/46. This revises our own earlier, smaller-sample finding that the SLA-only reward's collapse-avoidance edge over a best-possible fixed ceiling was clearly significant ($p = 0.0149$) – at full power that edge holds for the QoE-aware reward, but not for the SLA-only one ($p = 1.0$, a genuine null result at this sample size, not merely an underpowered one). The control loop adds negligible latency and model overhead, supporting the feasibility of real-time deployment. An offline evaluation under congested, randomised multi-slice traffic showed the learned policy can prioritise URLLC consistent with our earlier simulated work, though not yet as a clear win once slice priorities are weighted; a smaller-scale live evaluation of the same checkpoints found no arm forced into that tradeoff, since this campaign's calibrated ceilings do not approach genuine physical scarcity at real hardware scale, though the QoE-aware reward still gave URLLC a materially better experience score throughout.

Our future work will (1) investigate why the SLA-only reward's collapse-avoidance edge did not clearly replicate at full statistical power while the QoE-aware reward's did, (2) recalibrate ceilings and traffic to attempt a live reproduction of genuine multi-slice physical scarcity, and (3) extend this single-agent, single-gNB formulation toward the multi-agent, multi-gNB setting proposed in our systematic review [3].

Acknowledgment

This research was supported by the Ministry of Higher Education (MOHE), Malaysia through the Fundamental Research Grant Scheme (FRGS/1/2024/ICT06/TAYLOR/02/1).

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